BEYOND CREST FACTOR: A FIVE-CORPUS AUDIT OF SIGNAL ASSOCIATIONS IN SPEECH ANTI-SPOOFING

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ABSTRACT

Speech anti-spoofing systems can obtain convincing benchmark scores while responding to corpus-specific acoustic cues rather than synthesis artifacts. We distinguish three propositions: that a waveform feature separates labels, associates with a detector score within label, and changes a score under an intervention passing prespecified audio-quality gates. The locked audit registers 28 descriptors on three waveform views, eight published detector score artifacts, three discovery corpora, and two held-out confirmation corpora. Across 6,720 within-class screen cells, the exploratory, discovery-frozen Spectra-AASIST/full-waveform/spoof-class crest-factor candidate is negative in all three discovery corpora but attenuates sharply. In In-the-Wild its adjusted association is small ($\rho = -0.0326$), while in ASVspoof5 it is null after the registered controls ($\rho = 0.0014$), despite a negative unadjusted association. Thus this candidate fails the fixed five-corpus portability condition. Nineteen other registered feature/view/class units meet the descriptive association rule, but are not promoted post hoc to paired interventions. The frozen crest/control panel also fails its 90% per-arm quality-retention gate before detector scoring. The result is a conservative negative portability finding for the frozen crest-factor candidate, together with a reproducible association atlas; it makes no causal shortcut claim.

Index Terms— speech deepfake detection, anti-spoofing, shortcut learning, evidence audit, robustness

1. INTRODUCTION

Speech anti-spoofing countermeasures must distinguish bona fide speech from synthetic, converted, or replayed speech. Modern systems can learn effective spectro-temporal representations; for example, AASIST uses integrated spectro-temporal graph attention [1]. However, strong scores do not by themselves establish that a system responds to the intended manipulation artifacts. Müller et al. found that silence duration can correlate with the target label in ASVspoof data and that this cue alone can yield strong discrimination [2]. More generally, controlled shortcut conditions can produce near-perfect or worse-than-chance countermeasures without a reliable signal of the intended task competence [3]. Recent cross-domain evidence further indicates that audio-deepfake generalization gaps can reflect distributional difference, rather than only a harder version of the same task [4].

The present study begins with a practical observation: an interpretable feature—including crest factor—may be label-discriminative, correlated with a detector score, or sensitive to a waveform-family transformation targeted at it; these are different claims. Confusing them risks attributing model reliance to a feature that merely tracks label prevalence, corpus identity, duration, or loudness. We therefore audit published per-sample scores without recomputing baseline inference and use a staged, locked decision process.

This paper makes two completed contributions and one explicitly conditional design contribution.

- It registers a cross-dataset, multi-model association design that separates label effects from within-class score-feature associations and fixes the acceptance criterion before confirmation data are analyzed.
- It reports the exploratory discovery-frozen crest slice's failed five-corpus portability test, preserves the separate 19-unit descriptive atlas without post-hoc intervention promotion, and records a stopped quality-gated crest/control panel before detector scoring.
- It provides a conditional mitigation protocol, not an evaluated method, that activates only after a future valid sensitivity result.

We report only stages that completed their predeclared evidence gates. Sections 3–4 distinguish locked methodology, three discovery screens, held-out confirmation, and a quality-gated intervention attempt.

2. BACKGROUND AND AUDIT TARGET

The ASVspoof 2019 challenge provides a common setting for logical-access spoofing research [5]. Let $y_i \in \{0, 1\}$ denote the registered label of utterance i (0 bona fide and 1 spoof), f_i an interpretable waveform feature, and $s_{m,i}$ the published score of model m. Each score artifact is oriented to increasing spoof evidence only after a label-direction check; artifacts that fail an exact sample-ID join or an orientation check are excluded rather than heuristically repaired. Let c_i contain registered nuisance variables such as duration, integrated loudness, and available speaker or attack metadata.

The first audit target is the within-class association $\rho_{m,y} = \text{Spearman}(f_i, s_{m,i} \mid y_i = y)$ and its rank-residual analogue after controlling for c_i. This target is not a causal estimand: a residual association may still be caused by unobserved corpus properties. A subsequent paired waveform-family transformation supplies the sensitivity target $\Delta s_{m,i} = s_m(T_\theta(x_i)) - s_m(x_i)$, subject to prespecified quality gates on intelligibility proxies, transcript agreement, loudness, and transform-induced clipping. Such a transformation is targeted at a cue but is not an isolated intervention on that cue. The distinction structures the three hypotheses below.

3. LOCKED STUDY DESIGN

3.1. Inputs, feature registry, and provenance

All datasets, model revisions, and published score artifacts are pinned in the repository manifest. Discovery uses ASVspoof 2019 LA, ASVspoof 2021 LA, and ASVspoof 2021 DF; In-the-Wild and ASVspoof 5 [6] are held out for confirmation. The 2021 challenge adds channel/compression variation and unmatched evaluation conditions, so these corpora are not treated as interchangeable replications of the 2019 setting [7]. The score panel consists of Spectra-AASIST, AASIST, Res2TCNGuard, RawTFNet, WhisperMFCCMesoNet, W2V2-AASIST, XLSR-SLS, and RawBMamba. Where a published score artifact is absent, the cell is reported as missing and is never imputed. This design uses the Arena score artifacts as published evidence, rather than spending additional compute to reproduce their baseline inference. Accordingly, its estimands concern pinned released score artifacts, not re-executed current implementations or independent architectural mechanisms. The accompanying artifact bundle preserves the pinned manifest, executable analysis, and compact hash-sealed results; raw corpora remain external to the repository. A citable archival locator will be added when disclosure policy permits.

The frozen v1_28 registry extracts 28 amplitude, spectral, excitation, phase, and modulation descriptors from full-waveform, deterministic-crop, and pre-emphasized views. Voice-dependent descriptors retain missing values for unvoiced or unsuitable signals; they are not converted to zero. For corpora above 50,000 examples, the analysis selects a deterministic, class-balanced subset using seed 2609 and at most 5,000 examples per class.

3.2. H1: association and portability

For each dataset, feature, and waveform view, H1 reports feature-label AUROC. For each model, dataset, class, feature, and waveform view, it reports within-class Spearman association and partial rank association. The latter is the correlation of rank residuals after an intercept, ranked duration and integrated loudness, and available speaker/attack indicators; a tested feature or score is removed from its own control set. P-values receive Benjamini–Hochberg correction across the complete 1,344-cell screen within each corpus. Candidate estimates from the frozen manifest receive 2,000 label-stratified clustered bootstrap replicates during held-out confirmation, using speaker ID when available and otherwise source utterance ID. Exclusions are limited to missing stable IDs, unreadable waveforms, required labels, or published scores; every count is preserved in the accompanying artifacts.

A feature/view/class unit is portable only if at least five of eight models each have the same direction in at least four of the five core datasets, are BH-significant in both held-out confirmation datasets, and match the held-out direction. This full model-level criterion is fixed before confirmation analysis. Crest factor is one candidate in this registry, not the pre-supposed conclusion.

3.3. H2: planned quality-gated paired interventions

The planned quality-gated H2 sensitivity study requires an H1-frozen candidate, 500 examples per class per core dataset, and a four-model panel. A future transformed example is retained only if STOI is at least 0.95, original-to-transformed transcript WER is at most 5%, loudness drift is at most 0.2 LU, and no transform-induced clipping occurs, and the registered target-feature direction holds; an arm is dropped below 90% retention. A future sensitivity claim also requires a nonzero bootstrap interval, standardized paired score change of at least 0.2, and consistent direction in at least three datasets and three of four models.

Only its score-blind exploratory component has completed: a 1,000-clip ASVspoof 2019 LA crest/control quality screen with DRC-3, DRC-6, +0.1-dB gain, and polarity. It stopped before detector scoring because three arms fail the locked retention gate. It is therefore neither a completed four-model, multi-dataset H2 nor

Table 1. Completed evidence and H2 quality-gate outcome. The five-corpus association outcome is not a causal result and does not select new H2 candidates.

Artifact	Observed status
H1 screens on three discovery plus two held-out corpora	6,720 within-class tests (5 corpora $\times$ 8 models $\times$ 2 classes $\times$ 3 views $\times$ 28 features), with exact joins for every model and corpus.
Fixed crest-factor slice: full waveform, spoof class	Spectra-AASIST partial ρ is -0.454, -0.099, and -0.065 on 2019 LA, 2021 LA, and 2021 DF. It is -0.0326 in In-the-Wild ($q = 0.000523$) but 0.0014 in ASVspoof5 ($q = 0.937089$) after registered adjustment; it therefore fails portability.
Registered five-corpus association rule	19/168 feature/view/class units meet the full fixed sign, held-out-confirmation, and $\geq$ 5/8-model subcriterion; none is added post hoc to H2.
Frozen H2 crest/control quality panel	DRC-3, DRC-6, and gain retain 18.1%, 0.6%, and 64.6%, respectively, below 90%; polarity alone retains 99.9%. No detector scores, paired deltas, or H3 training are produced.

evidence of isolated crest-factor causality. The currently instantiated crest follow-up is exploratory because its discovery-to-H2 candidate rule was introduced during the discovery loop.

3.4. H3: conditional feature-conditioned mitigation

H3 is not run. It is confirmatory only after an H2 sensitivity result, at which point an equivalent-recipe baseline and feature-conditioned head would be compared under fixed BF16 seeds, throughput settings, and held-out EER gates. This conditional design does not assume that feature conditioning is beneficial [8].

4. RESULTS

Table 1 records the completed evidence and H2 quality-gate outcome, and Fig. 1 makes the frozen crest slice and the quality-gate stop directly inspectable. The 19 fixed-rule registry entries are preserved as a complete descriptive table, not a post-hoc intervention selector. No H2 delta, EER, or causal claim is reported. Eighteen of the 19 units are bona-fide-class associations; the remaining spoof-class unit is pre-emphasized spectral flatness. The registry also includes a separate bona-fide/full-waveform crest-factor association, which is distinct from the frozen spoof-class crest candidate and is not reused for H2.

5. LIMITATIONS AND RESPONSIBLE REPORTING

The proposed audit cannot prove that every unmeasured cue is irrelevant, and its interventions cover only registered transformations that pass automated quality gates. Metadata availability differs by corpus, so the partial analyses must report which controls are present rather than implying uniform adjustment. Exact corpus–class coverage and the held-out bootstrap cluster context are preserved in the accompanying hash-bound reproducibility material. Published score artifacts broaden architecture coverage but do not replace access to all model internals. The conditional mitigation can establish only a bounded robustness result under the registered splits and transformations, not universal safety against new generators or post-processing pipelines.

The H1 matrix now contains two held-out corpora, which expose why a repeated discovery direction alone is insufficient: the frozen Spectra-AASIST/full-waveform/spoof crest candidate is null after adjustment in ASVspoof5. The 19 registry-wide associations are descriptive screen results, so they neither identify a causal mechanism nor authorize post-hoc waveform transformations. The H2 crest run was frozen before those confirmation outcomes, but three of its four arms fail the 90% retention gate before detector scoring. It therefore supplies no paired-sensitivity estimate.

6. CONCLUSION

We present a reproducible association-portability audit and a path toward a future test of detector sensitivity: exact artifact joins, within-class adjusted associations, quality-gated paired interventions, and a conditional robustness test. Five corpora show that the exploratory, discovery-frozen Spectra-AASIST/full-waveform/spoof crest-factor direction does not survive the registered portability condition. The result also exposes a broader association atlas without converting it into post-hoc interventions. The pre-frozen crest transformations fail their own quality gate before score measurement. The paper therefore makes no causal feature-reliance claim.

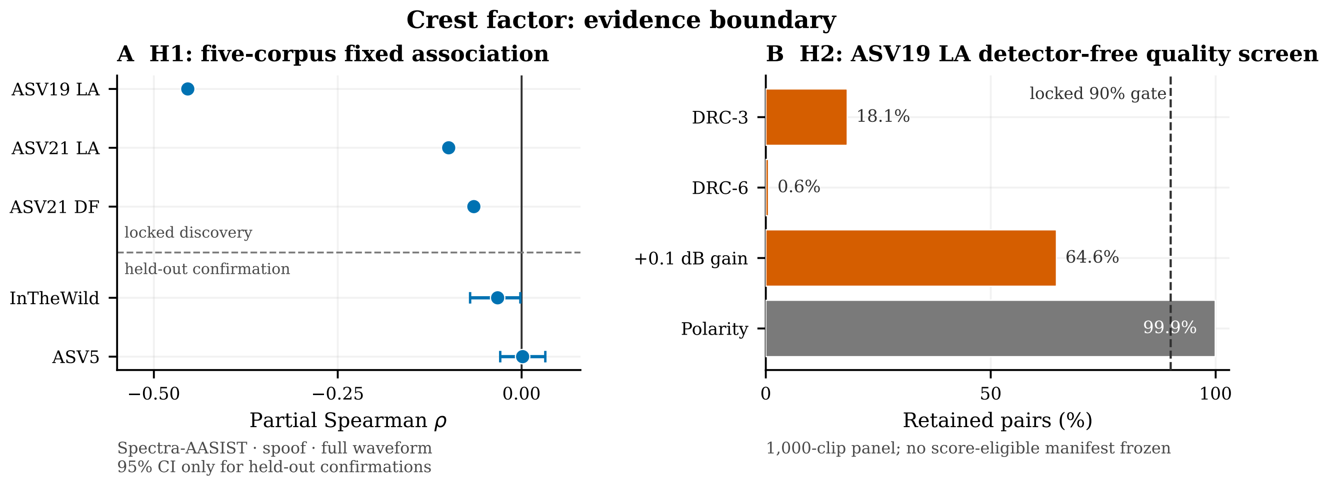

Fig. 1. Evidence boundary for the discovery-frozen Spectra-AASIST/full-waveform/spoof crest-factor slice. (A) Fixed-analysis adjusted associations across the fixed five-corpus order, separated into locked discovery and held-out confirmation; 95% clustered intervals appear only for the two sealed held-out confirmations. The slice fails the paper's fixed model-level portability condition. (B) Detector-free H2 retention for four fixed crest/control arms on the 1,000-clip ASVspoof2019 LA quality panel against the locked 90% gate. Polarity alone clears the per-arm rate, but three arms fall below it, so no score-eligible retained-pair manifest was frozen and no detector scoring was performed. The panels are descriptive displays only; no cross-panel relationship, pooled estimate, ranking, new test, or causal inference is computed.

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